

Concurrent Feedback VR Rhythmic Coordination Training

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ABSTRACT

Action Observation VR tools have had observable success in teaching a wide variety of skills. The authors' previous development includes a first-person VR tool designed for users to learn drumming exercises and improve rhythm via HMD. This contribution is an extension that implements concurrent feedback and multimodal cues to further immerse users in a coherent experience. Another virtual scene is included to help users improve their polyrhythms, a historically challenging musical concept to learn through traditional methods. The aim of this work is to use synchronized multimodal techniques with realtime feedback capabilities to develop a new drumming practice tool and to devise a potentially more effective, method to practice polyrhythms.

Index Terms: H.5.1 [Multimedia Information Systems]: Artificial, augmented, and virtual realities; H.5.5 [Sound and Music Computing]: Systems; H.5.m [Information Interfaces and Presentation]: Miscellaneous

1 INTRODUCTION

The practice of drumming and expressing rhythm, like any motor skill, requires practice before successful execution [6]. It entails both gross and fine motor skills as small muscles in the fingers are used for stick control while movement of larger muscles such as those in arms and torso are stretched to reach various instruments in the drummer's kit. Equally important is the ability for drummers to entrain to a musical piece's time, to play "in the pocket."

Action Observation (AO) is an application of the phenomenon in which observing the behaviour of another person or avatar produces similar neural activity as that performed by oneself [10]. When AO first-person virtual reality (VR) is part of the learning or practicing phase, benefits have been observed in tools that teach skills ranging from prosthetic limb control [10] to Tai Chi [4].

These findings influenced the authors' previous work, where three different AO first-person VR perspectives were utilized in the learning phase of a tool teaching rhythm [8]. These perspectives included Photospherical Video, Programmed Animation, and Mocap Animation, and all were developed by the Unity Game engine and deployed to the Meta Quest 2 head-mounted display (HMD).

In the learning phase, users could mimic the movement of each perspective's avatar's drum sticks to learn rhythms before removing the HMD and recording newly learned rhythms. While learning via AO, users could strike pads with real drum sticks or simply "air drum" while their view of one of the three virtual scene perspectives would simultaneously display the position of their hands using the HMD's frontal cameras.

Of the three perspectives, this current contribution focuses on extending functionality of the Programmed Animation perspective to include concurrent feedback, functionality to teach polyrhythms, and multimodal cues.

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Figure 1: User wearing an HMD to learn and practice via first-person Action Observation

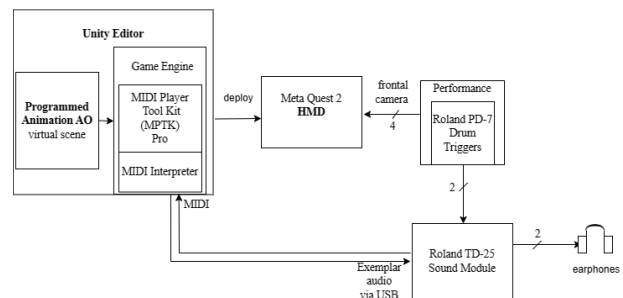


Figure 2: Block diagram for the system

2 FEEDBACK FUNCTIONALITY

The Programmed Animation perspective uses Unity's built-in Animator to control two drum stick objects to create a visual scene of drumming without any timing intricacies resulting from a human performance [8]. This set-up also simplifies the process of creating new exercises, as timing of left and right stick animations can be sequenced into new rhythms, bypassing the need for new recordings. Since this perspective creates a metronomically superior performance compared to that of the Photospherical Video or Mocap Animation perspectives, it was the best fit for a concurrent feedback extension.

Concurrent feedback is feedback that occurs at the same time as execution of a skill. Concurrent feedback on general human motion is not common, as standard pedagogy for physical movement typically includes feedback only after completion of the motion being taught [4]. However, training systems utilizing VR can provide concurrent feedback immersively through display systems such as HMDs [4] and have been shown to be helpful in terms of skill acquisition for novice learners [9].

Concurrent feedback was implemented through the use of a Unity scene using Maestro MIDI Player Tool Kit (MPTK) Pro, a Roland V-drums TD-25 sound module, and Roland PD-7 pads, as seen in Figure 2. The sound module connects to the PC running the Unity

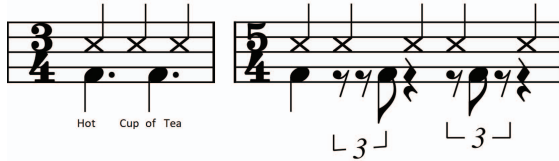


Figure 3: 3:4 and 4:5 polyrhythms

scene and receives the Unity scene's avatar's live audio feed via USB connection. The sound module also outputs audio triggered by the drum pads. Users monitor their performance synchronously with that of the avatar's via stereo earphones connected to the module.

A script within the Unity scene handles the drum pads' MIDI data and compares these events with the avatar performance via the Time class in Unity. Thresholds are set to determine whether a hit is early, late, or on time within some tight tolerance. This is then reported to the user via simple text or through multimedia means such as light objects or spatial audio expression within the virtual scene.

3 POLYRHYTHM TRAINING

A separate extension involving polyrhythms was also developed with the Programmed Animation perspective. A polyrhythm is a relatively easy to understand but relatively difficult to perform musical technique. It consists of the simultaneous expression of multiple mathematically related subdivisions of time by an individual or an ensemble. The mathematical relationship of a polyrhythm is $x:y$ where x and y are nonequal integers of positive value not equal to 1. x may be referred to as the basic pulse, and y is often referred to as the counter rhythm [5]. Therefore, 2:1 and 3:1 are not polyrhythms, but 3:2, 2:3, and 4:5 are instances of a virtually endless class of juxtaposed divisions that can be called polyrhythms.

Despite relatively simple-looking mathematical relationships, even basic polyrhythms such as 3:2 can be difficult for non-expert musicians to perform. Furthermore, listeners with musical experience may detect timing imperfections in the performance of certain polyrhythms with relative ease [2], which demands particularly stringent execution of polyrhythms.

Learning new exercises or pieces that are polyrhythmic is often difficult, demanding that the player listen and perform with respect to two tempos. However, the brain can only focus on one tempo at a time [3]. This presents challenges in the practice room that are somewhat unique considering technical practice of high-proficiency musicians. Specifically, while slowing down the tempo is a widely used technique that works on most practice material [1], it is not so effective while learning a polyrhythm. Alternative methods for polyrhythmic learning involves metric patterns in language [7]. For example, the way the phrase "hot cup of tea" is naturally spoken by most English speakers rhythmically represents the 3:2 polyrhythm [7]. However, these too are limited, as there does not exist a helpful phrase for every polyrhythm. For example, as shown in Figure 3, a 5:3 polyrhythm, like most complex polyrhythms, lacks a helpful verbal equivalent.

As seen in Figure 4, users can select the tempo of the basic pulse through a slider on the upper portion of the UI, and the counter rhythm adjusts with respect to the slider and the option selected via the dropdown menu. The elements of the polyrhythm are played on different sources, and the concurrent feedback GameObjects and text can be set with respect to either the basic pulse or counter rhythm.

4 CONCLUSIONS

This work expands on previous development through the inclusion of concurrent feedback and polyrhythmic capabilities, expanding

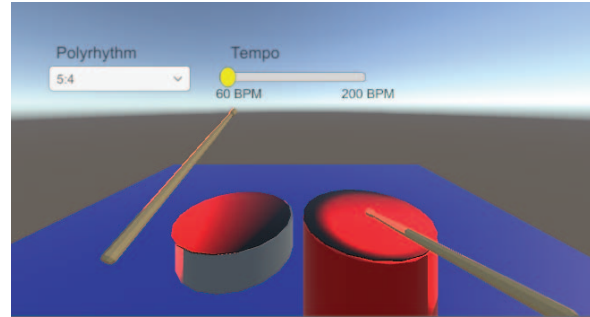


Figure 4: Programmed Animation perspective with polyrhythmic functionality and concurrent feedback. The red spot light GameObjects indicate the current performance is ahead of the beat and the user should slow down to play on beat.

the tool's potential to beginner and advanced use cases. In the near future we would like to further improve the user experience before conducting an experiment with test subjects to validate pedagogical effectiveness. Continuous concurrent feedback has been observed to help in certain skill acquisition applications [9], and it will be valuable to observe results of this approach in a musical setting.

Future work of interest include Meta Quest 2's hand-tracking 2.0 functionality for smoother hand visualizations in VR, utilization of the Meta Quest Pro (for AR modes and video see-through capabilities), extension to include both legs and feet to allow for multipart polyrhythms, inclusion of haptic feedback for complimentary gestalt cues, and expansion of exercises to include drumming with brushes rather than just sticks.

REFERENCES

- [1] K. A. Ericsson, R. R. Hoffman, A. Kozbelt, and A. M. Williams. *The Cambridge Handbook of Expertise and Expert Performance*. Cambridge University Press, 2018.
- [2] B. C. Fidalì, È. Poudrier, and B. H. Repp. Detecting perturbations in polyrhythms: effects of complexity and attentional strategies. *Psychological research*, 77(2):183–195, 2013.
- [3] M. Frishkopf. West African Polyrhythm: culture, theory, and graphical representation. *ETLTC: 3rd ACM Chapter Conf. on Educational Technology, Language and Technical Communication*, Jan. 2021. doi: 10.1051/shsconf/202110205001
- [4] T. N. Hoang, M. Reinoso, F. Vetere, and E. Tanin. Onebody: remote posture guidance system using first person view in virtual environment. In *Proc. Nordic Conf. on Human-Computer Interaction*, pp. 1–10, 2016. doi: 10.1145/2971485.2971521
- [5] P. Magadini. *Polyrhythms: the musician's guide*. Hal Leonard Corporation, 2001.
- [6] R. A. Magill and T. D. Lee. *Motor learning: concepts and applications: laboratory manual*. WCB McGraw-Hill, 1998.
- [7] A. Neely. 7:11 Polyrhythms. Youtube, 2019.
- [8] J. Pinkl and M. Cohen. Design of a VR Action Observation Tool for Rhythmic Coordination Training. In *IEEE Conf. on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 762–763, 2022. doi: 10.1109/VRW55335.2022.00232
- [9] R. A. Schmidt and G. Wulf. Continuous concurrent feedback degrades skill learning: Implications for training and simulation. *Human Factors*, 39(4):509–525, 1997.
- [10] M. Yoshimura, H. Kurumadani, J. Hirata, H. Osaka, K. Senoo, S. Date, A. Ueda, Y. Ishii, S. Kinoshita, K. Hanayama, and T. Suragawa. Virtual reality-based action observation facilitates the acquisition of body-powered prosthetic control skills. *J. of NeuroEngineering and Rehabilitation*, 17(1):113, 2020. doi: 10.21203/rs.3.rs-16403/v1